Matching chess.com's accuracy

One round of hill-climbing a game review's accuracy, held out by game

Chess Coach research lab

Abstract

The coaching app's game review gives each side an accuracy, and the player reads it beside the one chess.com gives the same game. The app's first instrument followed Lichess's published method: a score for each move from the share of winning chances it gives away, then a game score that averages a volatility-weighted mean and a harmonic mean of those scores. On 60 of the player's games, 30 of them held out, it missed chess.com's number by 7.5 points on average, no better than guessing 73.8 for every side (7.9). One change, a plain mean of per-move scores on a steeper per-move curve, cut the held-out error to 4.9 points (a paired change of -2.6 [-4.8, -0.9]) and raised the correlation with chess.com's number from 0.49 to 0.76. Merged with the app's grading of moves an endgame tablebase decides, it is the instrument the app ships (5.0). The lab recomputes every number from the recorded per-game results, and reruns all three instruments on engine evidence it rebuilt on the app as it now runs: no held-out error moves by more than 0.17 points.

1 The number the player compares

After a game, chess.com shows each side an accuracy between 0 and 100, and so does the app's game review. The player reads the two side by side, so a review whose accuracy disagrees with chess.com's by a wide margin for no visible reason costs it the player's trust. chess.com's formula is not published, and its number is not a ground truth about how well anyone played. The goal here is narrower: parity with the number the player sees.

2 Data and protocol

Games. The app's archive holds 192 of the player's games with chess.com's accuracy for both sides, recorded when the daily routine fetched each game. 60 were sampled, stratified by length (40 plies or fewer, 41 to 80, more than 80) and by result; the other 132 form a dev pool. Each sampled game counts twice, once for each side.

Split. The 60 games were split 30 for train and 30 for test, at random within each length bucket from a seeded generator, with the pilot games read while building the eval kept in train. Seeds were tried in order until the baseline's test-minus-train difference in both error and bias had a 95% interval containing zero, so that the split itself did not favour one half; 1 seed was rejected. Work on a round saw only the train games and the dev pool. The held-out games were scored once per round, for its decision.

Evidence. The app reviewed each game through its own entry point at two engine budgets, and the documents were frozen: *routine* (a sweep of 1,000,000 nodes a position, every flagged move verified at 750,000 nodes and 3 lines, as the daily routine reviews a game), which gives the headline numbers, and *phone* (a sweep of 100,000 nodes, with verification for at most 25% of the moves, as the app reviews a game on a phone), which is the guardrail. The engine is deterministic at a fixed node count on one thread, so one run is the whole measurement, and an instrument is measured by reading the frozen evidence again under its code.

Statistics. The headline is the mean absolute error (MAE) between the app's accuracy and chess.com's, over sides; within 5 is the share of sides within 5 points, the bias is the mean signed error, and r is Pearson's correlation over sides. Every interval is a 95% bootstrap over games (a game's two sides share its evidence), and a variant's difference from the baseline is paired on the same resampled games.

3 The instruments

Every instrument scores each graded move (book and forced moves are left out, and so are moves without an exact evaluation, except in v2 those a tablebase decides) from Δ, the share of the mover's winning chances the move gives away against the engine's best, in percentage points:

$$s(\Delta) \;=\; \min\Big(100,\; \max\Big(0,\; 103.1668\, e^{-d\,\Delta} - 3.1669 + 1\Big)\Big)\,.$$

- **The baseline** is Lichess's published method: $d = 0.0435$, and a side's accuracy is the average of two means of its scores, one weighted by volatility (the standard deviation of the winning chances over a sliding window of positions, one position for every 10 of the side's graded moves and between 2 and 8 wide, each weight held between 0.5 and 12) and one harmonic.
- **Round 1** keeps the curve's shape with a steeper decay, $d = 0.14$, and a side's accuracy is the plain mean of its scores.
- **The app (v2)** is round 1 over the moves the app now grades: since a later change, a move an endgame tablebase decides is graded by its exact outcome.

A move that gives away 2 points of winning chances scores 92 on the baseline's curve and 76 on the steep one; 5 points, 81 and 49; 10 points, 65 and 23 (Figure 1).

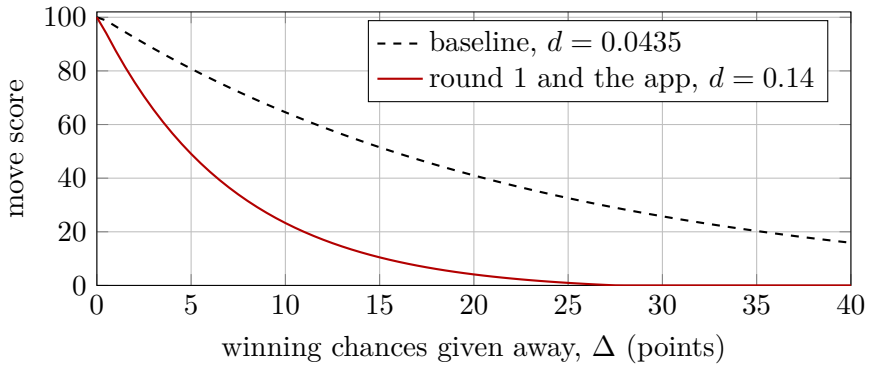
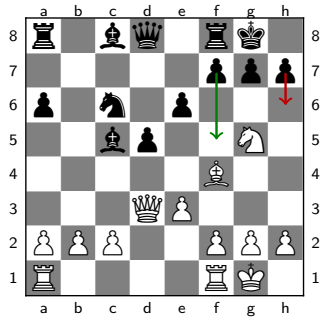


Figure 1: The per-move score on each curve, sampled from each instrument's own code.

4 Why the baseline missed

The baseline barely carried information about chess.com's number. Guessing the train sides' mean chess.com accuracy, 73.8, for every side scores 7.9 on test and 7.2 on train, against the baseline's 7.5 and 8.3. The error was not a constant offset (the baseline's bias on test is −0.5) but a tilt with the game's length (Table 1): short games read about 7.7 points below chess.com, long games about 5.1 above.

Both halves of the baseline's average explain the short end. In a short game a single blunder is a large share of the side's moves; its score near zero sinks the harmonic mean, and its window has the game's largest swing, so the volatility weights count it most. Figure 2 shows the held-out side the round improved most: Black gives away little until 13. . . h6, which allows 14. ♕h7#. Its score of 1.3 pulls the harmonic mean to 12.3 and the weighted mean to 46.8, and the side reads 29.6 against chess.com's 78.9. The plain mean of the same moves on the steep curve reads 75.5.



The game: 1. d4 d5 2. ♗f4 c5 3. ♘f3 e6 4. e3 ♘c6 5. ♗b5 a6 6. ♗a4 b5 7. ♘c3 bxa4 8. ♘xa4 ♘f6 9. dxc5 ♘e4 10. O-O ♘xc5 11. ♘xc5 ♗xc5 12. ♘g5 O-O 13. ♕d3 h6 14. ♕h7#
Played: 13. . . h6 (red); the engine's best was 13. . . f5 (green).
UCI: `h7h6`; best `f7f5`
FEN: `r1bq1rk1/5ppp/p1n1p3/2bp2N1/5B2/3QP3/PPP2PPP/R4RK1 b - - 3 13`

Black's graded move	win% lost	Lichess curve	steep curve
3. . . e6	0.4	99.4	95.8
4. . . ♘c6	0.2	100.0	98.4
5. . . a6	2.1	91.9	74.6
6. . . b5	0.0	100.0	100.0
7. . . bxa4	0.3	99.5	96.3
8. . . ♘f6	2.9	88.8	66.6
9. . . ♘e4	6.4	75.9	39.9
10. . . ♘xc5	0.0	100.0	100.0
11. . . ♗xc5	0.0	100.0	100.0
12. . . O-O	3.8	85.4	58.7
13. . . h6	78.1	1.3	0.0
the side's accuracy		29.6	75.5

Figure 2: The held-out side whose error round 1 shrank most, from the evidence the lab rebuilt: Black's 11 graded moves in a 27-ply game, the move played at the worst one (red) and the engine's best (green). chess.com gives the side 78.9.

length	sides	baseline	round 1	the app (v2)
short (40 plies or fewer)	24	−7.7	−4.1	−4.1
medium (41 to 80)	66	−2.4	+0.7	+0.7
long (over 80)	30	+5.1	+3.5	+3.9

Table 1: Mean signed error (the app's accuracy less chess.com's) by the game's length, over all 60 games' sides, routine evidence.

5 Round 1: a plain mean on a steep curve

Thesis. chess.com's accuracy is close to an average of per-move scores, each on a curve much steeper than Lichess's. The round's notes give the grounds: chess.com's description of its method as mostly averaging, and the example points of an independent public replication. A plain mean cannot be sunk by one move the way the harmonic mean is, and the steeper curve makes up for it by charging small concessions more.

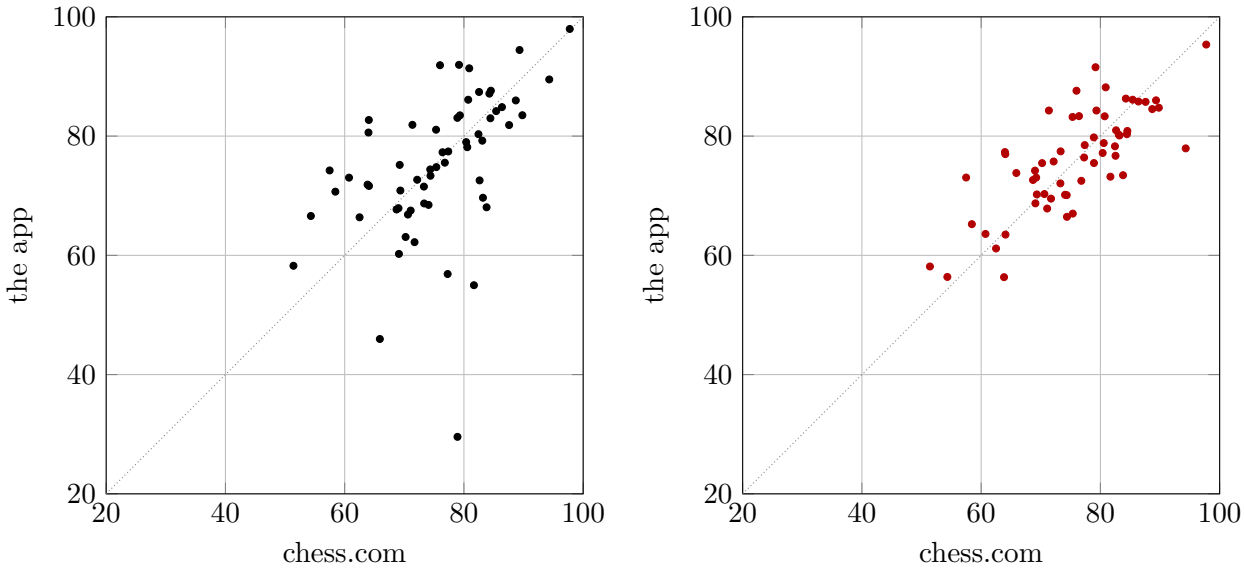
Before the held-out games. On the dev pool (132 games, never part of the sample) the change took the MAE from 8.0 to 4.3 (paired 95% interval [−4.90, −2.61]) and r from 0.61 to 0.86. Decays from 0.12 to 0.16 landed within 0.6 points of each other, so the gain is the shape, not a tuned constant.

Result. On the held-out games the MAE fell from 7.5 [5.3, 10.0] to 4.9 [3.7, 6.2], a paired change of −2.6 [−4.8, −0.9] points, and r rose from 0.49 [0.29, 0.73] to 0.76 [0.61, 0.88] (Table 2, Figure 3). Train improved with it (8.3 to 3.6), and so did the phone guardrail (7.5 to 5.4 on test). The length tilt shrinks but does not close: short games still read about 4.1 points low and long ones 3.9 high.

instrument	test MAE	train MAE	within 5	bias	r	vs baseline
routine evidence (the headline)						
baseline (Lichess)	7.5 [5.3, 10.0]	8.3	50%	−0.5 [−4.1, 2.8]	0.49 [0.29, 0.73]	–
round 1	4.9 [3.7, 6.2]	3.6	63%	+0.5 [−1.6, 2.6]	0.76 [0.61, 0.88]	−2.6 [−4.8, −0.9]
the app (v2)	5.0 [3.8, 6.3]	3.6	63%	+0.6 [−1.5, 2.7]	0.76 [0.61, 0.88]	−2.5 [−4.7, −0.7]
phone evidence (the guardrail)						
baseline (Lichess)	7.5 [5.5, 9.8]	8.1	48%	+0.2 [−3.3, 3.3]	0.50 [0.30, 0.74]	–
round 1	5.4 [4.0, 6.8]	4.6	57%	+0.0 [−2.3, 2.4]	0.72 [0.54, 0.86]	−2.1 [−4.1, −0.5]
the app (v2)	5.4 [4.0, 6.8]	4.6	57%	+0.0 [−2.3, 2.4]	0.72 [0.54, 0.86]	−2.1 [−4.1, −0.5]

Table 2: Each instrument against chess.com, with 95% bootstrap intervals over games. Every column but the train MAE is on the held-out games. MAE and bias in accuracy points, the app's less chess.com's; the last column is the change in test MAE from the baseline, paired on the same games.

Merged with tablebase grading. The app had meanwhile begun grading moves a tablebase decides by their exact outcome. Round 1's formula over that graded set is the app's instrument, v2, and costs nothing measurable: 5.0 on test (paired change from the baseline −2.5 [−4.7, −0.7]), 3.6 on train and 5.4 on phone evidence.



(a) Baseline: MAE 7.5, $r = 0.49$.

(b) The app (v2): MAE 5.0, $r = 0.76$.

Figure 3: The 60 held-out sides, the app's accuracy against chess.com's, routine evidence. The dotted line is parity.

What was tried and not run as rounds. The hill-climb stopped after round 1 because no candidate beat round 1 by more than 0.5 on the dev pool while improving train. These were tried on the dev pool and train and not run as rounds:

- book and forced moves counted: inside the noise or worse on the dev pool and train.
- flatter cp-to-win curve for accuracy: inside the noise or worse on the dev pool and train.
- floors and caps on per-move scores: 6.4 on dev, worse than round 1.
- book moves as best on top of round 1: cut the length tilt, MAE inside noise.
- per-category move scores (bands, classes, blunder cap): dev 4.46 / 4.50 / 4.30 against round 1's 4.29 in-sample; cross-validated 4.6.

6 Moving the history

The archive keeps every game's accuracy as the routine computed it, so a trend that mixed old numbers with new ones would show a jump that is only the change of instrument. The app therefore carries a stored accuracy only when it was taken under the current instrument, and otherwise reads the game again from its stored per-move evaluations. That is safe only if those evaluations give back the stored numbers under the old instrument, and the check of that was fixed before it ran: a whole-game mean gap of at most 0.5 points and a 90th percentile of at most 1.5. It held when recorded (0.34 and 1.04 over 454 sides) and holds on the app's archive as it is now (0.38 and 1.16 over 466 sides, the largest gap 7.0). Since the first run the archive has gained 42 games stored under later instruments, which the old instrument cannot reproduce and the check leaves out, as its definition says. The older rows, stored with no instrument named, come back within 0.03 points on average (286 sides); the later rows named after the old instrument come back within 0.92 (180 sides). The rule does not separate the two, and why the named rows reproduce less closely is not established here.

7 Reproduction in the lab

Recomputed. The lab recomputes every aggregate from the per-game results the records keep (`summarize.mjs`), and they match the published ones to the character. The records are read at `b3b827d`.

Rerun. The engine evidence the hill-climb used was lost, so the lab rebuilt it for the same games on the app as it now runs (`d8b7712`) and read it under each instrument (`reproduce.mjs`). The app's review pipeline changed after the hill-climb, so the evidence differs in detail, and the two older instruments needed one adjustment: they take the root position's winning chances as an argument the current app no longer passes, so they run with the app's document builder passing it again, as the app of their day did. Without it the baseline's volatility weights fail on a game whose first moves are graded, and such a game reads far below its recorded accuracy. With it, no held-out MAE moves by more than 0.17 points (Table 3).

		test MAE		train MAE		test r	
evidence	instrument	recorded	rerun	recorded	rerun	recorded	rerun
routine	baseline	7.5	7.5	8.3	8.3	0.49	0.49
routine	round 1	4.9	5.0	3.6	3.7	0.76	0.76
routine	the app (v2)	5.0	5.0	3.6	3.6	0.76	0.76
phone	baseline	7.5	7.6	8.1	8.2	0.50	0.51
phone	round 1	5.4	5.5	4.6	4.5	0.72	0.71
phone	the app (v2)	5.4	5.5	4.6	4.5	0.72	0.71

Table 3: The recorded numbers beside the lab's rerun on rebuilt evidence.

8 Limitations

- chess.com's formula is unknown, and parity with it is the goal, not a measure of play.
- 60 games of one player, rated 948–1167 on chess.com in these games. The held-out half keeps the claim honest about this player, not about others.
- The tilt with length remains: short games read low and long games high under every variant tried.
- The held-out games were scored for each round's decision. With one round kept and the others stopped on the dev pool, the room for a lucky pick is small, but not zero.